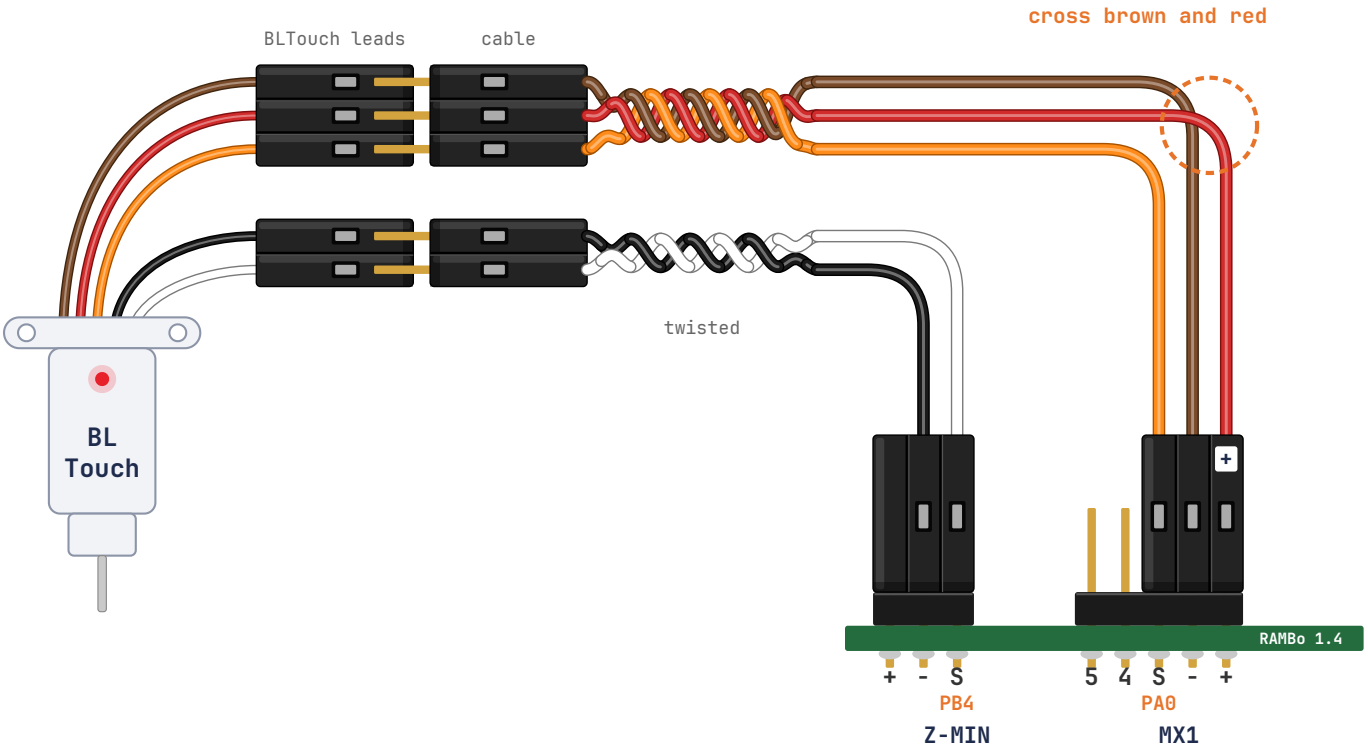
LULZBOT TAZ 6 · RAMBO 1.4 · KLIPPER

BLTouch cable

One cable from the BLTouch to two headers on the RAMBo. The three control wires go to **MX1**. The two probe wires go to **Z-MIN**, in the place of the washer wire.

Cross the brown and red wires

The plug of the BLTouch has GND, then 5 V. The MX1 header has 5 V, then GND. A straight 3-wire cable connects 5 V to GND. The probe before this one drew 1.39 A from the 5 V rail of the RAMBo. Test the cable with a meter before you connect the probe (page 5).

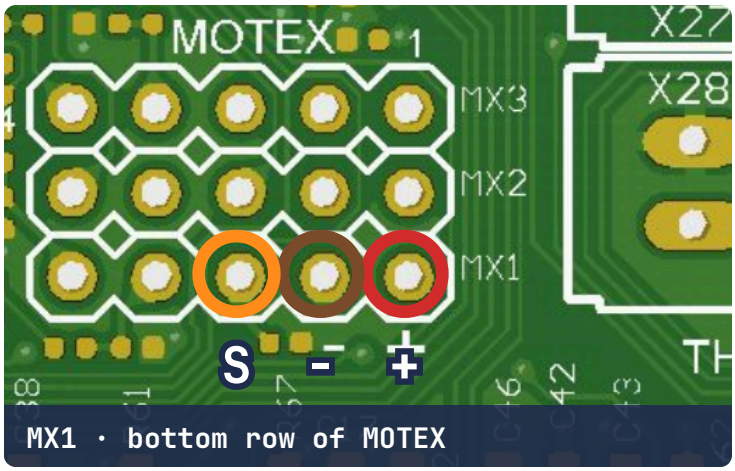


Side view. The plugs at the RAMBo end are down on their pins. The board marks and the Klipper pins are under each pin.

WIRE	BLTOUCH FUNCTION	RAMBO HEADER	PIN	KLIPPER PIN
Brown	GND	MX1	2 (−)	
Red	+5 V	MX1	1 (+)	
Orange	Control (servo)	MX1	3	`PA0`
Black	GND	Z-MIN	− (middle)	
White	Probe signal	Z-MIN	S	`PB4`
None		Z-MIN	+ (5 V)	
None		MX1	4 and 5	`PC5`, `PL5`

The two headers on the RAMBo

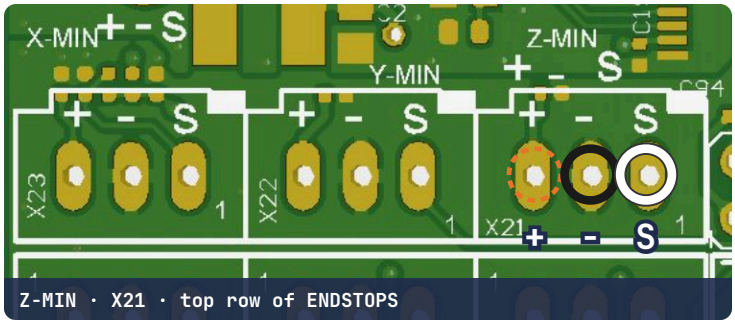
The marks are from the silkscreen of the RAMBo 1.4a. Use the marks on the board to find each pin, because the board can be in the control box in any direction.



MX1 · bottom row of MOTEX

MX1: control wires

- MOTEX is a block of three rows of five pins, next to the thermistor headers. MX3 is the top row and **MX1 is the bottom row**.
- Pin 1 is at the end with the **1** and **+** marks. Pin 2 has the **−** mark.
- Pin 1 is 5 V. Pin 2 is GND. Pin 3 is PA0, through a 120 Ω filter.
- Pins 4 and 5 (PC5, PL5) stay empty.
- If the holes of MOTEX have no pins, solder a 1×3 straight pin header into pins 1 to 3 of MX1.



Z-MIN · X21 · top row of ENDSTOPS

Z-MIN: probe wires

- The MIN headers are the top row of ENDSTOPS: X-MIN, Y-MIN, then **Z-MIN** (X21).
- The marks are **+ − S**. S is pin 1. The washer wire is in this header now.
- S goes to PB4. The board has a 10 kΩ pull-up to 5 V, a 120 Ω filter and a 100 pF capacitor on it.
- The + pin (5 V) stays empty. The BLTouch gets its power from MX1.

Source: UltiMachine RAMBo 1.4a design files, github.com/ultimachine/RAMBo-1.4. Sheet 2, Motor Extensions (X20 = MX1). Sheet 3, Endstops (X21 = Z-MIN). The pictures are crops of the 3D print of the top of the board.

Make the cable

Parts

5	Stranded wires, 24 or 26 AWG, in brown, red, orange, black and white. The length is in step 1.
1	Dupont 1×3 housing and 1 Dupont 1×2 housing, for the probe end.
5	Dupont crimp pins (male), for the probe end. Look at the BLTouch leads first. If they end in pins, use sockets here.
1	Dupont 1×3 housing and 3 crimp sockets (female), for MX1.
1	3-position housing and 2 crimp terminals that fit Z-MIN. Use the plug of the washer wire as the sample. I did not check the part number of this housing.
	Heat-shrink tube, braided sleeve (optional), zip ties, a label for each board end.

Tools

- Dupont crimp tool and a wire stripper.
- Multimeter, with continuity and DC volts.
- Heat gun or lighter for the heat-shrink tube.

Steps

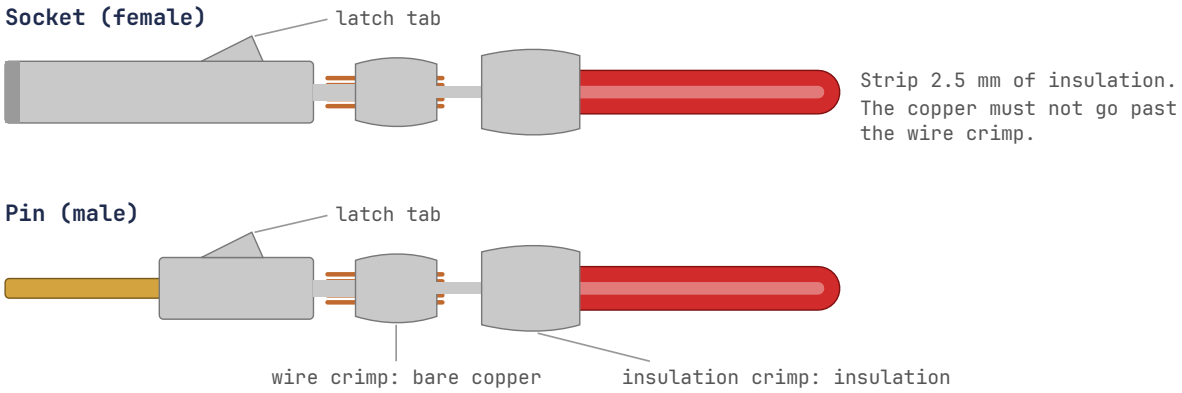
1. **Measure the path.** Move the toolhead to the end of X and to the top of Z. Lay a string from the BLTouch mount along the toolhead cable to the RAMBo. Add 200 mm to that length.
2. **Cut five wires** to that length.
3. **Twist the wires in two groups.** Twist black and white together, about one turn each 25 mm. Twist brown, red and orange together. The twist decreases noise on the probe signal.
4. **Probe end.** Crimp the pins. Put brown, red, orange in the 1×3 housing and black, white in the 1×2 housing, in the same order as the BLTouch plugs. Push each pair together and make sure that each color meets the same color.
5. **MX1 end.** Crimp the sockets. Put **red, brown, orange** in the 1×3 housing. Mark the red end of the housing **+**.
6. **Z-MIN end.** Crimp the terminals. Put them in the 3-position housing as **empty, black, white**, for **+ − S**. Compare with the washer plug.
7. **Finish.** Put a label on each board end: "MX1 +" and "Z-MIN". Shrink the tube on the ends, and put on the sleeve.

Route the cable

- Tie the cable to the toolhead cable each 100 mm. Keep a loop at the toolhead.
- Keep the cable away from the heater wires where you can.
- Before the last ties, move X and Z through their full travel by hand. Make sure that the cable does not pull or catch.

The connectors

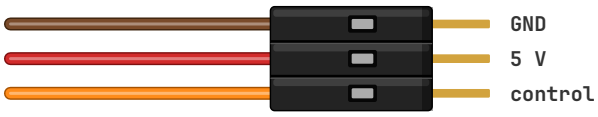
All the plugs are Dupont housings with 2.54 mm pitch. Each wire gets a crimp terminal. The terminal locks in the housing with its latch tab.



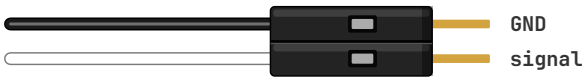
- Crimp the wire crimp first, then the insulation crimp. Pull on the wire: it must not move.
- Push the terminal into the housing with the latch tab on the side of the window, until it clicks.
- To remove a terminal, lift the window tab with a pin, then pull the wire.

The four finished ends

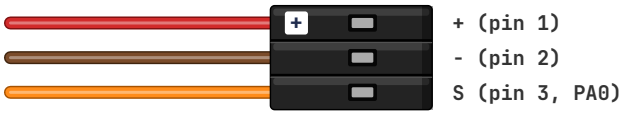
Each end is shown with the latch windows up and the mating face on the right. Compare your cable with these before you connect it.



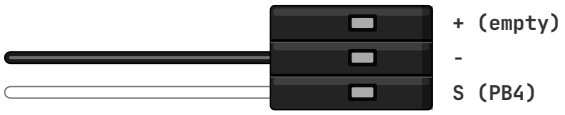
Probe end, 1×3 · to the BLTouch control plug



Probe end, 1×2 · to the BLTouch probe plug



MX1 end, 1×3 · red and brown crossed · + mark on the red end



Z-MIN end, 3-position · the + position stays empty

The Z-MIN housing must fit the header like the washer plug. If the washer plug is not a Dupont housing, use a housing of the same type as the washer plug.

Test before the probe goes on

1. **Continuity.** With the power off, check each wire from end to end. Then check each pair of wires next to each other in each plug: the meter must show an open circuit.

2. **The probe.** With the probe off the cable, measure the resistance from red to brown on the BLTouch. A reading of a few ohms or less is a short: do not use the probe. I do not have the normal value for this probe.

3. **Connect the board end.** With the power off, take the washer plug out of Z-MIN. Push the cable into Z-MIN and MX1, with the + mark on the + pin.

4. **Measure the voltage at the probe end.** Turn on the power, with the probe still off the cable. Put the red lead of the meter on the red wire and the black lead on the brown wire of the 1×3 plug.
   - **+5.0 V:** correct.
   - **−5.0 V:** red and brown are the wrong way round. Turn off the power and correct the MX1 plug.
   - **0 V:** the plug is on the wrong pins, or MX1 has no power. Do not continue.

   Then measure from white to black on the 1×2 plug. The pull-up on the board gives about 5 V.

5. **Connect the probe.** Turn off the power, connect the BLTouch, then turn on the power.
   - **Correct:** the pin goes down and up two times, then the light stays on. The LCD stays bright.
   - **Stop:** the LCD goes dim, the probe does not light, or the probe gets warm. Turn off the power immediately.

Then the config

Tell Claude that the probe passed. The `[bltouch]` section replaces the washer `[probe]`. The points of `[bed_tilt]` and the PRINT_START macro change too. The config is saved only when you say go, and the first test is a self-test with no movement.

```
[bltouch]
sensor_pin: ^PB4
control_pin: PA0
x_offset: ...     # measure from the nozzle to the pin
y_offset: ...
#z_offset: set with PROBE_CALIBRATE
```